The Concept of Novelty Detection Applied to Power Quality

Leandro Rodrigues Manso Silva
Eder Barbosa Kapish
Carlos Augusto Duque
Luciano Manhes de Andrade Filho
Augusto Santiago Cerqueira
Federal University of Juiz de Fora
Minas Gerais - Brazil

Paulo Fernando Ribeiro
Federal University of Itajuba
Minas Gerais - Brazil

***Abstract*—This paper presents a study related to a possible topology of a novelty detector to be applied in Power Systems applications. With the new scenario of Smart Grids a novelty detector should not be focused in the typical PQ disturbances but in any new deviation/variation that may happen in the signal. In this way, the traditional methodology of novelty detection, that is based on a training stage is not suitable. The proposed detection methodology divides the signal in fixed length frames and is based on the comparison of the current frame with the reference one due to an specific metric. Results for two scenarios will be shown and a Hit Rate up to 99.0% with a False Alarm 1.0% were obtained.**

I. Introduction

An important issue related to Power Quality (PQ) events is the identification of the disturbance source. To do this identification, the engineer depends on recorded data captured by monitoring equipment scattered throughout the system. To storage the data efficiently, the instrument must be equipped with signal processing techniques in order to detect the relevant data that must be stored for future analysis.

In the new smart grids scenario the detection algorithm must be capable of looking for any new kind of disturbances that may occur in these systems. So, the conventional detectors, focused on detecting only the classical PQ disturbances (Sags, Swell, Harmonics, Transients) may not perform very well. In this way, the Novelty concept [1] fits very well in being the basis of the detection algorithm, since it looks for any new "abnormality" present in the signal.

The traditional novelty detection concept, much used in biomedical area [2], [3]; in industrial failure detection [4]; video surveillance [5], [6]; sensor networks [7]; and other applications is slightly different of the one used in this paper. The traditional novelty detection is used when a very large number of examples of "normal" condition is available and where there are insufficient data to describe "abnormalities", and in this way, it is based on a training stage [8].

This paper proposes a methodology that divides the signal into fixed size frames and uses the novelty detection concept to signalize when a signal frame is different of a reference one related to some metric and its application is as follows: (i) the signal is segmented in fixed length frames; (ii) the current frame is compared with a reference one; and (iii) if they were different according with some metric and a threshold, a novelty is detected. It differs from the classical novelty detection since a training set is not used.

The paper is organized as follows. Section II presents the proposed novelty detection scheme. Section III presents the obtained results, comparing three detection methods. Section IV shows the effect Novelty Detector when it is used as a basis of a Power Quality Data Recorder. And, finally, Section V presents the conclusions.

II. The novelty detector

The main objective of the detector is to detect events that are truly novelties, ignoring steady state signals with or without frequency variations. The proposed Novelty Detector is composed by two detection branches as can be seen from Fig. 1.

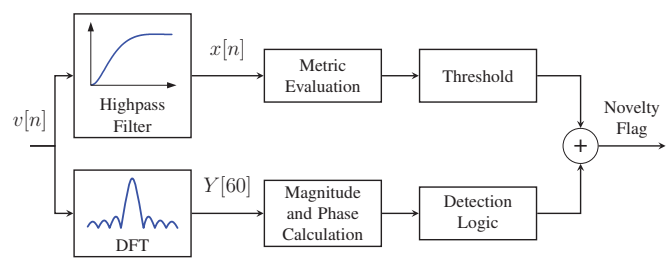

Fig. 1. Block Diagram of the proposed novelty detector.

The first branch is composed by a high-pass filter, and a metric evaluation followed by a thresholding process. This branch is responsible to effectively detect the novelty frames. The detection is built on comparing the current frame with a reference one according to some metrics. The reference frame is the last detected novelty frame. Three metrics were used in order to test which one performs better, they are: The Difference of Frame Energies (DFE), the Energy of the Residue (ER), and the Mathematical Morphology (MM). The filter has the aim of attenuating the effect of the fundamental component and low frequency harmonics, in order to increase the detection resolution. Several filters were tested such as a

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notch filter centered in 60 Hz, a DFT filter, also centered in 60 Hz and a 10th order high-pass filter with cutoff frequency equals to 720 Hz, and the conclusion was that the high-pass filter performed better than the others. The three metrics used in this works are described below.

1) The Difference of Frame Energies (DFE): The DFE is calculated as the absolute value of the subtraction of the current frame energy from the energy of the reference frame, as shown in (1). If DFE is higher than a given threshold a novelty frame is detected, and this frame is labeled as the new reference frame.

$$DFE = |EF_c - EF_r| \tag{1}$$

The energy of each frame is calculated according to the following equation:

$$EF = \sum_{n=0}^{M-1} |x[n]|^2 \tag{2}$$

where $x[n]$ is the n-th sample of the frame, and M is the number of samples per frame. It is important to note that $x[n]$ is the output of the highpass filter shown in Fig. 1.

2) The energy of the residue (ER): The energy of the residue is computed according with (3). The residue is the difference between the samples of the current frame with the corresponding samples of the reference frame.

$$ER = \sum_{n=0}^{M-1} |x_c[n] - x_r[n]|^2 \tag{3}$$

where, $x_c[n]$ is the sample of the current frame and $x_r[n]$ is the sample of the reference frame.

3) Mathematical Morphology (MM): Mathematical morphology operations [9] are used for signal processing based on the shape of the signal. The output sample of the mathematical morphology operations depends on the corresponding input sample and on the samples in its neighborhood. The size and shape of considered neighborhood are determined by a function s called structuring element.

The implemented methodology for the sake of comparison is based on [10] and uses the closing operation [11]. This closing operation of a function u is defined as follows:

$$u \bullet s = (u \oplus s) \ominus s \tag{4}$$

where, the $\oplus$ indicates the dilation operation (5) and the $\ominus$ indicates the erosion operation (6), both using the same function s.

$$(u \oplus s)[n] = \max\{u[n-m] + s[m]\} \tag{5}$$

where, $m \in S$; $n - m \in U$.

$$(u \ominus s)[n] = \min\{u[n+m] - s[m]\} \tag{6}$$

where, $m \in S$; $n + m \in U$.

In equations (5) and (6), U and S are the domains of definition of the functions u and s respectively. As u is a one dimensional signal, the function s is a binary vector, and in this paper, s is a unitary vector with size equals to 32 samples.

The closing operation is applied to the input signal $v[n]$ generating $v_M[n]$ that represents the shape of the input signal. Then, the MM metric is the energy of the difference of $v_{Mc}[n]$ of the current frame and $v_{Mr}[n]$ of the reference frame as shown below:

$$MM = \sum_{n=0}^{M-1} |v_{Mc}[n] - v_{Mr}[n]|^2 \tag{7}$$

where $v_{Mc}[n]$ and $v_{Mr}[n]$ are the shape of the current and reference frames, respectively.

The second branch of the detector comprises a logic to improve the performance of the detector. It is composed by a Discrete Fourier Transform (DFT) centered in 60 Hz, producing an output named Y_{60}, for each cycle of the signal. The magnitude ($|Y_{60}|$) and phase ($\angle Y_{60}$) of this complex value are calculated and used in the detection logic.

It is known that disturbances with frequencies near to the fundamental causes an error in the DFT estimation of this component, and the detection is based on this fact. Then, there are two main objectives for this second branch: detect low frequency disturbances, such as Flicker and Sub-Harmonics, and signalize if the signal presents frequency variations, in order to suppress false alarm detections.

The low frequency disturbances detection is based on the fact that they cause a significant error in the $|Y_{60}|$ estimation, while $\angle Y_{60}$ almost does not change. On the other hand, in frequency variation scenarios the phase of $Y[60]$ varies as well as its magnitude.

The value of the magnitude of the current cycle is subtracted from the magnitude of the reference cycle, generating an error signal, and the same is done for the phase. The detection is performed based on these two error signals, and an event is detected when the magnitude error is higher than a specified threshold and the phase error is near zero. The reference cycle is updated when an event is detected.

In Section III the comparison of the detector performance using the three described metrics will be shown for various scenarios.

III. Results

The signals used in the tests that were done to verify the Novelty Detector performance were divided in two classes: the signal class that comprises only steady state signals, such as the fundamental component and stationary harmonics; and the disturbance class comprising PQ disturbances such as: Oscillatory Transients, Time Varying Harmonics, Notches, Sags/Swells and Flicker.

The novelty detector was tested in two scenarios, each one with various SNRs (Signal to Noise Ratio). In both scenarios the disturbance class was generate containing one frame of

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fixed frequency sinusoid and the other with a disturbance (Sag, Swell, Transients, Harmonics or Notches). The signal class differ from one scenario to the other: in the first one, the signal is composed by two frames of fixed frequency sinusoids, and in the second, by two frames of frequency variation sinusoids with or without harmonic distortion. In both scenarios the signals are corrupted by noise.

The results obtained using only the first branch of the detector were compared in terms of its Receiving Operation Characteristics (ROC) curves and the optimum threshold value was chosen by means of the Sum-Product (SP) index [12] given by Eq. 8.

$$SP = \sqrt{\sqrt{S \times E} \times \frac{S+E}{2}} \tag{8}$$

where, S is the detector sensibility, defined as the probability of detection, and E is the specificity, defined as $(1 - P_{FA})$, where P_{FA} is the false alarm probability of the detector. The threshold value that maximizes the SP index is the optimum one.

This results are shown in Fig. 2 for the first scenario (without frequency variation signals) and in Fig. 3 for the second one (with frequency variation signals).

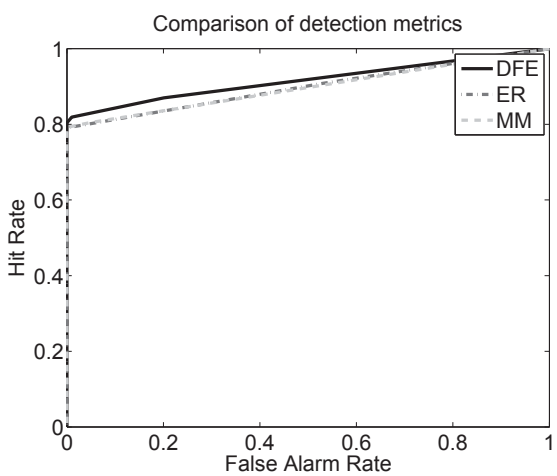

Fig. 2. Comparison of the novelty detection metrics.

The comparison of the detectors performance for the first scenario and a SNR of 30 dB is shown in Fig. 2. For higher SNRs all the metrics perform very well. It can be noted that the DFE metric performs better than the others. Using the optimum threshold, a Hit Rate of 81.4% and a False Alarm Rate of 1.6% are achieved for the detector that uses the DFE metric.

The comparison of the detection metrics for the frequency variation scenario is shown in Fig. 3. In this scenario only the detector that uses the DFE metric performs well. Using the same methodology to calculate the optimum threshold value, a Hit Rate of 78.6% and a False Alarm Rate of 2.0% were obtained.

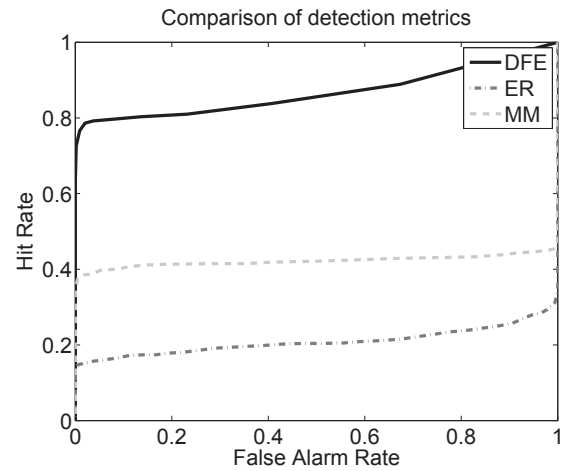

Fig. 3. Comparison of the novelty detection metrics.

In order to improve the performance of the detector, the second branch was added with two main objectives: improve the detection of the low frequency disturbances such Flicker and suppress the False Alarm detections due to frequency variant signals. The results using the complete detector are shown in Fig. 4 for the first scenario (without frequency variation signals) and in Fig. 5 for the second one (with frequency variation signals). These results are also for a SNR of 30 dB.

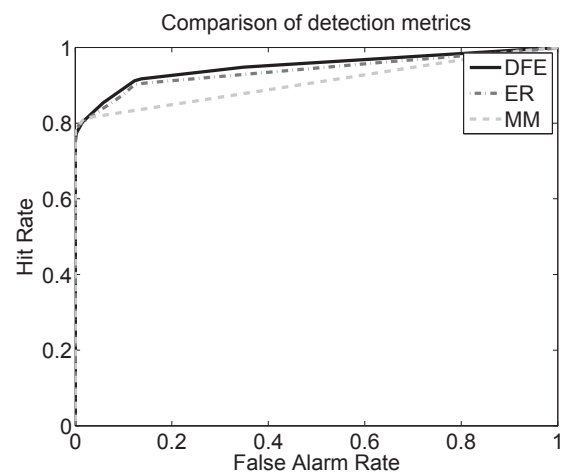

Fig. 4. Comparison of the novelty detection metrics.

It can be noted from Fig. 4 that the DFE detector is slightly better than the other, and using the optimum threshold value a Hit Rate of 85.4% and a False Alarm Rate of 5.8% were achieved. These results are better than the results obtained using only the first branch since the Hit Rate was increased. Note that the False Alarm Ratio was also increased, but it is not a big problem since the priority is detecting the novelties.

The results shown in Fig. 5, present a huge difference compared with the results of Fig. 2. This is due to the ability of the second branch of the detector in recognizing frequency variant signals, suppressing the detections due to

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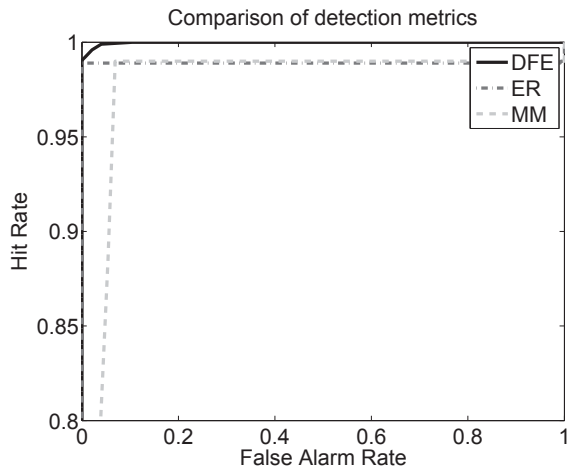

Fig. 5. Comparison of the novelty detection metrics.

this phenomena and increasing the detector resolution. A Hit Rate of 99.0% and False Alarm Rate of 1.0% were achieved using the optimum threshold value. It can be noted that, again, the DFE detector performed better than the others.

IV. The Impact of the Novelty Detection in a Power Quality Data Recorder

A continuous raw data recording of electrical signals is not a simple task since a large amount of data needs to be stored or transferred. Few commercial types of equipment are currently available that aims at recording continuous raw data at a high sampling rate [13]. For an efficient data storage, a detection technique is necessary, besides a good compression methodology.

Most of the conventional recorders are application oriented and are used only for acquiring either a short-term of abnormal signal [14], [15]. An efficient continuous power quality data recorder must be capable of detecting only the relevant data that needs to be stored, in order to store long periods of the signal in a small memory space. In this way, the use of a novelty detector as a basis for a storage system affects directly the amount of data that needs to be stored. Results showing the file size generated by a Power Quality Data Recorder with and without the use of the Novelty Detector are presented in Table I.

The signals shown in Table I have about 10 minutes of duration and each of them is composed by a combination of successive electrical disturbances. The disturbances for each case are:

- **S1:** sag, swell, odd harmonics, even harmonics , inter-harmonics and ramp magnitude variation;
- **S2:** ramp magnitude variation;
- **S3:** odd harmonics, even harmonics and odd and even harmonics simultaneously;
- **S4:** interahmonics;
- **S5:** ramp magnitude variation;
- **S6:** notches;
- **S7:** ramp magnitude variation.

The Table I presents some interesting results, for example: the second column presents the percentage of frames that were detected as a novelty frames during the tested interval; the third and fourth columns present the size needed to store the signals when the Novelty Detector is used and when it is not used, receptively; and the fifth column shows the Compression Rate (CR) that is the relationship between the numbers in the third and fourth columns.

It can be noted that the use of the Novelty Detector contributes to the efficiency of the data storage, decreasing significantly the amount of data that needs to be stored. This fact can be confirmed through the obtained CRs that are higher as the percentage of detected novelties is lower.

In the all above signal the system frequency is kept constant, however the proposed method can deal with time varying frequency being able to reconstruct the whole signal using an off line procedure that takes into account the frequency variation information and the shape of novelty frames. An illustrative example that highlights this characteristics is shown in Fig. 6 and in Fig. 7. The considered signal is compose by a simple sinusoid component with time varying frequency. In Fig. 6 the detector is sensitive to frequency variation (for example the high pass filter and the lower branch of the block diagram 1 were taken off). The thick blue line represents the novelty frames that were detected and the thin blue corresponds to the off-line reconstructed frames. As can be seen in this case a lot of novelty frames were detected.

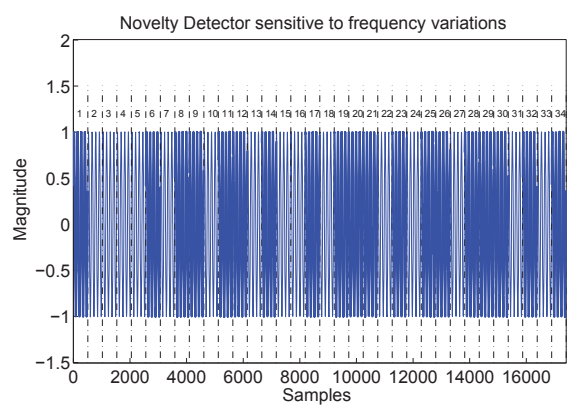

Fig. 6. Example using the Novelty Detector sensitive to frequency variation.

In Fig. 7, the results are shown for the case where the high pass filter and the lower branch of the block diagram 1 were used, and then, the detector is insensitive to frequency variation. Note that only a few novelties frames were detected.

Higher compression rate can be achieved by extra processing of the novelties frames such redundancy reduction through wavelet transforms followed by a lossless compression algorithm. These further signal processing is not object of the present paper but is interesting highlight that when they are used together with the present proposed a final compression rate of 1000:1 can be obtained.

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TABLE I
COMPARISON OF FILE SIZES GENERATED BY A POWER QUALITY DATA RECORDER WITH AND WITHOUT THE USE OF THE NOVELTY DETECTOR.

Signal	Percentage of Frames Detected as Novelties	File Size Without Novelty Detector	File Size With Novelty Detector	Compression Rate
S1	11.6 %	3256 kB	519 kB	6:1
S2	11.2 %	2315 kB	275 kB	8:1
S3	1.2 %	2693 kB	56 kB	48:1
S4	29.0 %	2832 kB	1008 kB	3:1
S5	2.5 %	2305 kB	91 kB	25:1
S6	1.2 %	1960 kB	40 kB	49:1
S7	3.9 %	2397 kB	136 kB	17:1

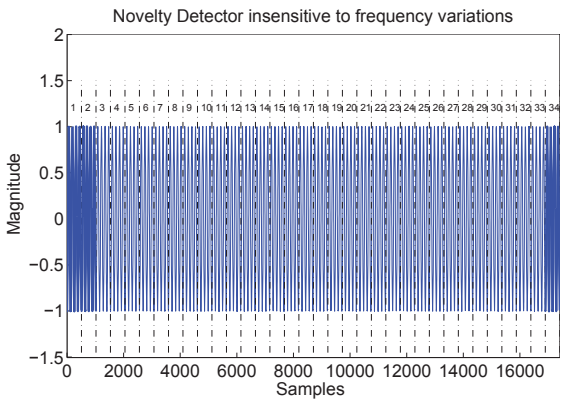

Fig. 7. Example using the Novelty Detector insensitive to frequency variation.

V. CONCLUSION

This paper presented a possible topology for a novelty detector applied in Power Systems applications within the new scenario of Smart Grids and focused in detecting PQ and other disturbances when any novelty or new deviation/variation occurs in the electrical signal.

With the results presented for the tested metrics, the DFE is more suitable to be used in the proposed Novelty Detector since it performed better in the tested scenarios, achieving high detection rates.

It was shown that the use of the Novelty Detector increases the performance a Power Quality Data Recorder since the amount of data that needs to be stored decreases significantly. This fact strengthens the applicability of the proposed methodology, particularly in the complex scenario and context of smart grids, due to its ability in detecting any variation in the system signal.

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